SQL • DATA ANALYTICS • PORTFOLIO PROJECT

ZOMATO

DATA ANALYSIS

SQL Project Report

Restaurant Data Exploration and Business Insights

PROJECT TYPE	**Personal Data Analytics Project**
DOMAIN	**Online Food Supply**
TECHNOLOGY	**MySQL | MySQL Workbench**
DATASET	**Zomato restaurant listings (approx. 9,551 records)**
PREPARED BY	**DHARANIDEVAN V**

Built with MySQL Workbench

1. Project Overview

The objective of this project is to explore a Zomato restaurant dataset using SQL, summarize restaurant characteristics, and identify patterns in country distribution, online delivery, ratings, votes, cuisines, table booking, and restaurant recommendations.

Objectives

- Inspect the dataset structure and check RestaurantID data quality.
- Create a country-name mapping using CountryCode.
- Review city-name spelling and encoding consistency.
- Use a window function for a rolling count.
- Calculate statistical summaries and rating categories.
- Analyse country distribution, online delivery, Indian restaurants, cities, cuisines, and table booking.
- Recommend moderately priced restaurants and identify a matching restaurant in Kolkata.

Tools and Dataset

- Database tool: MySQL / MySQL Workbench.
- Database and table used: zomato_s_a.zomato_dataset.
- Main fields used include RestaurantID, RestaurantName, CountryCode, Country_Name, City, Locality, Cuisines, Currency, Rating, Votes, Price range, Average Cost for two, Has Table booking, and Has Online delivery.

Note: Column names with spaces are enclosed in backticks in SQL. The exact field names should match the imported table in MySQL Workbench.

2. Data Exploration

2.1 Check Table Details

Use DESCRIBE to inspect the column names and data types:

```
DESCRIBE zomato_s_a.zomato_dataset;
```

To inspect the table definition and constraints:

```
SHOW CREATE TABLE zomato_s_a.zomato_dataset;
```

The query should be run in the user's database to capture the exact structure and constraints.

2.2 Duplicate Check in RestaurantID

```
SELECT
    RestaurantID,
    COUNT(*) AS ID_Count
FROM zomato_s_a.zomato_dataset
GROUP BY RestaurantID
HAVING COUNT(*) > 1;
```

Output: the query returned 0 rows.

RestaurantID	ID_Count
(no rows returned)	-

Interpretation: no RestaurantID appears more than once. All 9,551 records are unique restaurants, so no duplicate rows needed to be removed.

2.3 Column Removal

Each column was compared with the project tasks. A column was marked for removal when no task uses it or when it repeats information that is already in another column. The Locality and LocalityVerbose columns were also checked with a count query:

```
SELECT
    COUNT(*) AS total_rows,
    COUNT(Locality) AS locality_count,
    COUNT(LocalityVerbose) AS locality_verbose_count,
    COUNT(DISTINCT Locality) AS unique_localities,
    COUNT(DISTINCT LocalityVerbose) AS unique_verbose_localities
FROM zomato_s_a.zomato_dataset;
```

Output:

total_rows	locality_count	locality_verbose_count	unique_localities	unique_verbose_localities
9551	9551	9551	1208	1265

Interpretation: Locality and LocalityVerbose are both filled in all 9,551 rows, so neither has missing values. LocalityVerbose has 1,265 distinct values against 1,208 for Locality. The difference reflects the city name being added to the locality text, which splits a locality that appears in more than one city into separate values. The City column already holds this information, so LocalityVerbose adds nothing new.

Columns removed and the reason:

Column removed	Reason
Address	No task uses the full street address. City and Locality are enough for the location analysis.
LocalityVerbose	Repeats Locality with the city name added. The City column already stores the city.
Switch_to_order_menu	Not used by any task in the project.
Is_delivering_now	Not used by any task, and it only shows the delivery status at one moment in time.

Columns needed by later tasks were kept, including CountryCode (needed for the country mapping), City, Locality, Cuisines, Currency, Rating, Votes, Price_range, Average_Cost_for_two, Has_Table_booking, and Has_Online_delivery.

```
ALTER TABLE zomato_s_a.zomato_dataset
    DROP COLUMN Address,
    DROP COLUMN LocalityVerbose,
    DROP COLUMN Switch_to_order_menu,
    DROP COLUMN Is_delivering_now;
```

Conclusion: removing these four columns made the table simpler without losing any information needed for the analysis tasks.

2.4 Country Mapping

A separate country lookup table was created and joined to the restaurant dataset.

```
CREATE TABLE zomato_s_a.country (
    CountryCode INT PRIMARY KEY,
    Country_Name VARCHAR(100)
```

```
);

ALTER TABLE zomato_s_a.zomato_dataset
ADD COLUMN Country_Name VARCHAR(100);

UPDATE zomato_s_a.zomato_dataset AS A
JOIN zomato_s_a.country AS B
    ON A.CountryCode = B.CountryCode
SET A.Country_Name = B.Country_Name;
```

A NULL check returned zero rows after the update. This indicates every dataset row had a Country_Name populated at that point, but the country lookup entries should still be checked for correct CountryCode-to-country mappings.

```
SELECT *
FROM zomato_s_a.zomato_dataset
WHERE Country_Name IS NULL;
```

2.5 Correct City Names / Encoding

Three checks were run on the City column to find misspelled or inconsistent city names: extra spaces, similar-sounding names, and special characters.

Check 1: Extra spaces (TRIM)

```
SELECT DISTINCT City
FROM zomato_s_a.zomato_dataset
WHERE City <> TRIM(City);
```

This check returned no rows, so no city name has extra spaces.

Check 2: Similar-sounding names (SOUNDEX)

```
SELECT
    SOUNDEX(City) AS sound_code,
    GROUP_CONCAT(DISTINCT City) AS variants,
    COUNT(DISTINCT City) AS n
FROM zomato_s_a.zomato_dataset
GROUP BY SOUNDEX(City)
HAVING COUNT(DISTINCT City) > 1;
```

sound_code	variants	n
P230	Pasay City, Pasig City	2
P524	Panchkula, Pensacola	2

Pasay City and Pasig City are two different cities in the Philippines. Panchkula is in India and Pensacola is in the USA. These are false matches, so no change was made.

Check 3: Special characters

```
SELECT DISTINCT City
FROM zomato_s_a.zomato_dataset
WHERE City REGEXP '[^A-Za-z .-]';
```

City	Meaning
Brasília	Correct spelling with an accent letter
São Paulo	Correct spelling with an accent letter
İstanbul	Correct spelling with a special Turkish letter

City	Meaning
Cedar Rapids/Iowa City	Correct listed name that contains a slash

In the exported check file, the accented letters appear as ? because they are outside the plain English (ASCII) character set. The stored values are correct: the rolling count output in section 2.6 shows Brasília, São Paulo, and İstanbul with the right spellings.

Conclusion: none of the three checks found a real misspelling. The names are valid, so no UPDATE statement was needed.

2.6 Rolling Count Analysis

```
SELECT
    City,
    COUNT(*) AS restaurants,
    SUM(COUNT(*)) OVER (
        ORDER BY City
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
    ) AS rolling_count
FROM zomato_s_a.zomato_dataset
GROUP BY City;
```

Output (selected rows from 141 cities, in alphabetical order of City):

City	restaurants	rolling_count
Abu Dhabi	20	20
Agra	20	40
Ahmedabad	21	61
Albany	20	81
Allahabad	20	101
New Delhi	5473	7963
Noida	1080	9043
Winchester Bay	1	9550
Yorkton	1	9551

The restaurants are counted for each city, and the window function adds them up city by city to give a running total. The last rolling_count value is 9,551, which equals the total number of rows in the table, so the query is correct. The data covers 141 cities, and New Delhi alone has 5,473 restaurants. This is a cumulative (running) count, not a moving-window count.

2.7 Statistical Summary

```
SELECT
    MIN(Votes) AS Min_Votes,
    MAX(Votes) AS Max_Votes,
    ROUND(AVG(Votes), 2) AS Avg_Votes,
    MIN(Rating) AS Min_Rating,
    MAX(Rating) AS Max_Rating,
    ROUND(AVG(Rating), 2) AS Avg_Rating
FROM zomato_s_a.zomato_dataset;

SELECT
    Currency,
    COUNT(*) AS Restaurant_Count
FROM zomato_s_a.zomato_dataset
GROUP BY Currency
```

```
ORDER BY Restaurant_Count DESC;
```

Measure	Observed result
Minimum votes	0
Maximum votes	10,934
Average votes	156.91
Minimum rating	1.0
Maximum rating	4.9
Average rating	2.89

Currency is categorical, so it was summarized using counts by currency rather than MIN/AVG/MAX.

Most frequent currency counts reported:

Currency	Restaurant count
Indian Rupees	8,652
Dollar	482
Pounds(?)	80
Brazilian Real	60
Emirati Diram	60
Rand	60
NewZealand($)	40
Turkish Lira	34
Botswana Pula	22
Indonesian Rupiah	21
Qatari Rial	20
Sri Lankan Rupee	20

Some currency labels show possible encoding or spelling inconsistencies (for example, Pounds(?)); verify the source values before final submission.

2.8 Rating Category Creation

```
ALTER TABLE zomato_s_a.zomato_dataset
ADD COLUMN Rating_Category VARCHAR(20);

UPDATE zomato_s_a.zomato_dataset
SET Rating_Category =
    CASE
        WHEN Rating IS NULL THEN 'not rated'
        WHEN Rating >= 4.5 THEN 'excellent'
        WHEN Rating >= 4.0 THEN 'very good'
        WHEN Rating >= 3.5 THEN 'good'
        WHEN Rating >= 2.5 THEN 'average'
        ELSE 'poor'
    END;

SELECT
    Rating_Category,
    COUNT(*) AS Restaurant_Count
FROM zomato_s_a.zomato_dataset
GROUP BY Rating_Category
ORDER BY Restaurant_Count DESC;
```

Output:

rating_category	Restaurants	Share of all restaurants
average	3,737	39.13%
poor	2,334	24.44%
good	2,100	21.99%
very good	1,079	11.30%
excellent	301	3.15%

The rating thresholds were chosen for this project: 4.5+ Excellent, 4.0–4.49 Very Good, 3.5–3.99 Good, 2.5–3.49 Average, below 2.5 Poor, and NULL Not Rated.

Finding: the five categories add up to 9,551 restaurants, so every restaurant was classified. The largest group is average (3,737 restaurants, 39.13%), followed by poor (2,334, 24.44%) and good (2,100, 21.99%). Only 1,380 restaurants (14.45%) are very good or excellent, and just 301 (3.15%) are excellent. Almost two thirds of the restaurants (63.56%) are average or poor, which matches the overall average rating of 2.89 in section 2.7.

3. Data Analysis

3.1 Country Distribution Analysis

```
SELECT
    Country_Name,
    COUNT(*) AS Total_Restaurants,
    ROUND(
        COUNT(*) * 100.0 /
        (SELECT COUNT(*)
         FROM zomato_s_a.zomato_dataset),
        2
    ) AS Country_Percentage
FROM zomato_s_a.zomato_dataset
GROUP BY Country_Name
ORDER BY Country_Percentage DESC;
```

Output:

Country_Name	Total_Restaurants	Country_Percentage
India	8,652	90.59%
United States	434	4.54%
United Kingdom	80	0.84%
Brazil	60	0.63%
United Arab Emirates	60	0.63%
South Africa	60	0.63%
New Zealand	40	0.42%
Turkey	34	0.36%
Australia	24	0.25%
Philippines	22	0.23%
Indonesia	21	0.22%
Singapore	20	0.21%
Qatar	20	0.21%
Sri Lanka	20	0.21%
Canada	4	0.04%

The 15 countries add up to 9,551 restaurants. India has about 91% of the data (8,652 restaurants), and the United States is second with 4.54%. The percentages add up to 100.01% only because each value is rounded to two decimals. Because India dominates the dataset, most results reflect Indian restaurants.

3.2 Online Delivery Insights

```
SELECT
    Country_Name,
    COUNT(*) AS Total_Restaurants,
    SUM(
        CASE
            WHEN Has_Online_delivery = 'Yes' THEN 1
            ELSE 0
        END
    ) AS Online_Delivery_Restaurants,
    ROUND(
        SUM(
            CASE
                WHEN Has_Online_delivery = 'Yes' THEN 1
                ELSE 0
            END
        ) * 100.0 / COUNT(*),
        2
    ) AS Online_Delivery_Percentage
FROM zomato_s_a.zomato_dataset
GROUP BY Country_Name
ORDER BY Online_Delivery_Percentage DESC;
```

Output:

Country_Name	Total_Restaurants	Online_Delivery_Restaurants	Online_Delivery_Percentage
United Arab Emirates	60	28	46.67%
India	8,652	2,423	28.01%
Philippines	22	0	0.00%
Brazil	60	0	0.00%
United States	434	0	0.00%
Australia	24	0	0.00%
Canada	4	0	0.00%
Singapore	20	0	0.00%
Indonesia	21	0	0.00%
New Zealand	40	0	0.00%
United Kingdom	80	0	0.00%
Qatar	20	0	0.00%
South Africa	60	0	0.00%
Sri Lanka	20	0	0.00%
Turkey	34	0	0.00%

Online delivery is most common in the United Arab Emirates (46.67%, 28 of 60 restaurants), followed by India (28.01%, 2,423 of 8,652 restaurants), a difference of 18.66 percentage points. The other 13 countries show 0%, so these two countries account for all 2,451 online delivery restaurants in the dataset (25.66% of all 9,551). India alone holds 2,423 of them (98.86%). The 0% values may reflect how the dataset was collected rather than a real absence of delivery in those countries, so the comparison reflects this dataset and should not be generalized to entire national restaurant markets.

3.3 Focus on Indian Restaurants

```
SELECT
    COUNT(*) AS Total_Restaurants,
```

```
    ROUND(AVG(Rating), 2) AS Average_Rating,
    MAX(Rating) AS Highest_Rating,
    MIN(Rating) AS Lowest_Rating,
    SUM(Votes) AS Total_Votes
FROM zomato_s_a.zomato_dataset
WHERE Country_Name = 'India';
```

Metric	Result
Total restaurants	8,652
Average rating	2.77
Highest rating	4.9
Lowest rating	1.0
Total votes	1,187,163

The dataset contained 8,652 Indian restaurant records, with an average rating of 2.77, a maximum rating of 4.9, a minimum rating of 1.0, and 1,187,163 recorded votes.

3.4 City Analysis – Popular Locations

```
SELECT
    City,
    COUNT(*) AS Total_Restaurants
FROM zomato_s_a.zomato_dataset
GROUP BY City
ORDER BY Total_Restaurants DESC
LIMIT 3;
```

Rank	City	Listed restaurants
1	New Delhi	5,473
2	Gurgaon	1,118
3	Noida	1,080

New Delhi had the highest listing count among the three cities shared, followed by Gurgaon and Noida.

3.5 Cuisine Insights – Connaught Place

```
SELECT
    Cuisines,
    COUNT(*) AS Restaurant_Count
FROM zomato_s_a.zomato_dataset
WHERE Locality LIKE '%Connaught Place%'
GROUP BY Cuisines
ORDER BY Restaurant_Count DESC
LIMIT 10;
```

Cuisine combination	Restaurant records
North Indian | Mughlai	7
Café	6
North Indian	6
North Indian | Chinese | Italian | Continental	4
Fast Food | Burger	3
North Indian | Chinese | Mughlai	2
North Indian | Continental | Italian	2
North Indian | Chinese	2
Chinese	2
Fast Food	2

The most frequent complete cuisine combination in the query result was North Indian | Mughlai, with 7 records. Cafe and North Indian each had 6. Because the query groups the full cuisine text, it counts combinations rather than counting each cuisine separately across all combinations.

3.6 Table Booking Impact Analysis – Connaught Place

```
SELECT
    Has_Table_booking AS Table_Booking,
    COUNT(*) AS Total_Restaurants,
    ROUND(AVG(Rating), 2) AS Average_Rating
FROM zomato_s_a.zomato_dataset
WHERE Locality LIKE '%Connaught Place%'
GROUP BY Has_Table_booking;
```

Table booking	Restaurant count	Average rating
Yes	57	3.82
No	68	3.61

The average rating for restaurants with table booking was 3.82 compared with 3.61 for those without. The difference is 0.21 points. This is an association in the dataset and does not prove that table booking causes higher ratings.

3.7 Restaurant Recommendations

The project brief does not define exact thresholds for 'moderately priced' or 'best'. For this analysis, the following working criteria were selected: Price range = 2, Rating >= 4.0, Votes >= 100, and at least one of table booking or online delivery available. These are analyst-selected thresholds, not values prescribed by the brief.

```
SELECT
    RestaurantName,
    City,
    Average_Cost_for_two,
    Rating,
    Votes,
    Has_Table_booking,
    Has_Online_delivery
FROM zomato_s_a.zomato_dataset
WHERE Price_range = 2
  AND Rating >= 4.0
  AND Votes >= 100
  AND (
      Has_Table_booking = 'Yes'
      OR Has_Online_delivery = 'Yes'
  )
ORDER BY Rating DESC, Votes DESC,
         Average_Cost_for_two ASC
LIMIT 10;
```

Output (top 10 restaurants under these criteria):

Restaurant	City	Cost for two	Rating	Votes	Table booking	Online delivery
Caterspoint	Gurgaon	500	4.9	223	No	Yes
CakeBee	Coimbatore	350	4.9	200	No	Yes
Truffles	Bangalore	800	4.7	9,667	No	Yes
Tapri Central	Jaipur	750	4.7	1,469	No	Yes
Coffee Cup	Secunderabad	800	4.6	1,408	Yes	No
India Restaurant	Kolkata	800	4.6	1,219	Yes	Yes
Spezia Bistro	New Delhi	900	4.6	1,071	Yes	Yes

Restaurant	City	Cost for two	Rating	Votes	Table booking	Online delivery
Twigly	Gurgaon	500	4.6	395	No	Yes
Joey's Pizza	Mumbai	800	4.5	2,662	No	Yes
Burgrill	Chandigarh	500	4.5	249	No	Yes

Finding: all ten restaurants are in India and have a rating of 4.5 or higher. Caterspoint (Gurgaon) and CakeBee (Coimbatore) share the top rating of 4.9, but with only 223 and 200 votes. Truffles (Bangalore) has by far the most votes (9,667) with a rating of 4.7, which makes its rating the most trusted. Nine of the ten restaurants offer online delivery, and three (Coffee Cup, India Restaurant, and Spezia Bistro) offer table booking.

Kolkata recommendation

```
SELECT
    RestaurantName,
    City,
    Average_Cost_for_two,
    Rating,
    Votes,
    Has_Table_booking,
    Has_Online_delivery
FROM zomato_s_a.zomato_dataset
WHERE City = 'Kolkata'
  AND Price_range = 2
  AND Rating >= 4.0
  AND Votes >= 100
  AND (
      Has_Table_booking = 'Yes'
      OR Has_Online_delivery = 'Yes'
  )
ORDER BY Rating DESC, Votes DESC,
         Average_Cost_for_two ASC
LIMIT 1;
```

Field	Result
Restaurant	India Restaurant
City	Kolkata
Average cost for two	800 currency units (dataset value)
Rating	4.6
Votes	1,219
Table booking	Yes
Online delivery	Yes

India Restaurant was the top result returned by the selected Kolkata criteria. The cost value is recorded as 800 in the dataset; the currency should be checked for the row before writing a currency symbol. This result is the top match under the chosen sorting and thresholds, not proof of an objectively best restaurant. It is also number 6 in the overall top 10 list above.

4. Overall Findings

- The RestaurantID duplicate check returned no rows, so all 9,551 records are unique.
- Four columns (Address, LocalityVerbose, Switch_to_order_menu, and Is_delivering_now) were removed because no task needed them or they repeated other columns.

- The city name checks (TRIM, SOUNDEX, and special characters) found no misspellings, so no corrections were needed.
- The rolling count over 141 cities ended at 9,551, which matches the total number of rows.
- India holds 90.59% of the restaurants (8,652 records), and the United States is second with 4.54%.
- The Indian subset contained 8,652 restaurant records and 1,187,163 recorded votes.
- Online delivery appears in only two countries in the data: the UAE (46.67%) and India (28.01%). The other 13 countries show 0%.
- New Delhi had the highest city listing count among the reported top three, with 5,473 records.
- North Indian | Mughlai was the most frequent complete cuisine combination in the Connaught Place result, with 7 records.
- Connaught Place restaurants with table booking had an average rating of 3.82 compared with 3.61 for restaurants without it.
- India Restaurant was the top Kolkata result returned under the selected moderate-price, rating, vote, and facility criteria.
- Most restaurants are rated average (39.13%) or poor (24.44%), and only 3.15% are excellent.
- Nine of the top 10 moderately priced restaurants offer online delivery, and the highest-voted one is Truffles in Bangalore with 9,667 votes.

5. Limitations and Validation Notes

- Confirm the exact spelling and data type of all column names in MySQL Workbench. Column names with spaces require backticks.
- Verify that Yes/No values in the facility columns are actually stored as text. If they are stored as 1/0, change the CASE and WHERE conditions accordingly.
- Check for NULL values in Rating, Votes, Average Cost for two, Price range, and country mapping before finalizing the report.
- The manually created country lookup should be checked for accurate mappings, not just non-NULL values.
- The cuisine query counts complete cuisine combinations, not each individual cuisine separately.
- Rating and votes are dataset indicators; they do not guarantee food quality or customer uniqueness.
- The 'moderate price' threshold and recommendation criteria were selected for this analysis because the project brief did not specify exact cut-offs.

6. Conclusion

This project used MySQL to explore a restaurant dataset and answer practical questions about restaurant distribution, online delivery, ratings, votes, cuisine listings, table booking, and restaurant recommendations. The analysis demonstrates the use of aggregation functions, CASE expressions, filtering, grouping, sorting, joins, and window functions. The data-quality checks listed above should be reviewed before final submission.

End of Report